

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0746

Shady Oaks Wind Farm 100 kV
March 22, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0746. The Interconnection Customer proposes a 222.4 MW natural gas combined cycle facility interconnecting at the Shady Oaks Wind Farm 100 kV substation in the Illinois area. The request was received on February 18, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 223 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,077,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Beaver 230 kV/1 kV XFMR: 100.8% loading (1,793 MVA) following the outage of the Eufaula to Beaver 500.0 kV; DFAX = 13.7%.

Beaver 500-1 kV Auto: 100.8% loading (1,793 MVA) following the outage of the Beaver to Eufaula 500.0 kV line; DFAX = 13.7%.

The Exira to Shady Oaks Wind Farm 100.0 kV Line is overloaded to 100.5% of its 165 MVA emergency rating for the loss of the Exira to Shady Oaks Wind Farm 100.0 kV Circuit 1. The project contributes 12.9% DFAX (28.7 MW) to this overload.

The Eufaula 230/1 kV TX is overloaded to 100.1% of its 1,453 MVA emergency rating for the loss of the Eufaula to Imperial 500.0 kV. The project contributes 12.3% DFAX (27.3 MW) to this overload.

For the loss of Imperial to Eufaula 500.0 kV L, the Eufaula 500/1kV Autotransformer loads to 100.1% of its 1,453 MVA rating. Project DFAX: 12.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beaver 230 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$16,659,000	4% (\$619,000)
Beaver 500-1 kV Auto	Replace transformer with higher-rated unit	\$30,625,000	4% (\$1,139,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Exira to Shady Oaks Wind Farm 100.0 kV Line	Reconductor line	\$1,104,000	100% (\$1,104,000)
Eufaula 230/1 kV TX	Replace transformer with higher-rated unit	\$11,866,000	4% (\$441,000)
Eufaula 500/1kV Autotransformer	Replace transformer with higher-rated unit	\$20,809,000	4% (\$774,000)
	Total allocated to project		\$4,077,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.